

# The sharp Berry–Esseen bound for independent arrays with small Lyapunov ratio

Logan Bell

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## Abstract

Every finite independent centered array with positive total variance and finite third absolute moments satisfies  $\Delta \leq C_E \ell$  whenever  $\ell \leq \exp(-\exp(20004))$ , where  $\Delta$  is the Kolmogorov distance of the standardized sum from the normal law,  $\ell$  is its Lyapunov ratio, and  $C_E = (3 + \sqrt{10})/(6\sqrt{2\pi})$ . The coefficient is optimal. The proof develops the variational, uniform-jitter, and local-comparison methods of He and Cheng for summands with different laws. A finite approximate-period argument and a quantitative form of Esseen’s moment inequality identify a common scale. Support variations put every coordinate of a maximizing array into two narrow clusters or a small neighborhood of zero. An exact comparison with a common-diameter Bernoulli array completes the proof. All parameters are explicit. The common-diameter bound and an auxiliary unrestricted bound are supplied by companion papers with validated finite calculations.

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## 1 The theorem and the proof

Let  $X_1, \dots, X_n$  be independent centered real random variables. Put

$$B^2 = \sum_i \mathbb{E}X_i^2 > 0, \quad T = \sum_i \mathbb{E}|X_i|^3 < \infty, \quad \ell = T/B^3, \quad (1.1)$$

and write

$$\Delta = \sup_x \left| \mathbb{P} \left\{ B^{-1} \sum_i X_i \leq x \right\} - \Phi(x) \right|, \quad \mathcal{R}(X_1, \dots, X_n) = \Delta/\ell.$$

Here  $\Phi$  and  $\phi$  are the standard normal distribution function and density. Zero coordinates are allowed. Set  $c_E = 3 + \sqrt{10}$  and  $C_E = c_E \phi(0)/6$ .

**Theorem 1.1.** *For every array satisfying (1.1),*

$$\ell \leq \ell_0 := \exp(-\exp(20004)) \implies \Delta \leq C_E \ell.$$

*The coefficient  $C_E$  cannot be decreased, even when all coordinates have the same two-point law.*

The value of  $\ell_0$  is a sufficient threshold. The proof does not estimate the largest admissible threshold, and the sharp inequality for unrestricted independent arrays remains open.

He and Cheng [4, Theorem 1.1, Sections 2–5, and Appendix A] prove the sharp inequality for iid sums once their sample size exceeds a universal threshold. Their proof supplies three mechanisms used here: coordinate contamination gives support equations, uniform jitter reveals a lattice scale, and a local comparison separates small and large residual variance. Independence with different laws requires additional estimates. We prove these for the original coordinates, including those outside the two-cluster class. In particular, conditioning on a Bernoulli count does not make the residuals independent; our conditional Fourier argument retains each label together with its residual.

Two finite bounds from the companion papers enter the proof:

$$\mathcal{R}(h(B_1 - p_1), \dots, h(B_m - p_m)) \leq C_E, \quad (1.2)$$

$$\mathcal{R}(X_1, \dots, X_n) < 9/20. \quad (1.3)$$

In (1.2), the  $B_i$  are independent Bernoulli variables with arbitrary parameters and positive total variance, and  $h > 0$ . This is the common-diameter theorem [1, Theorem 1.2(ii)]; its proof uses

Schulz's binomial theorem [6, Theorem 1] and validated finite calculations. Equation (1.3) follows from the unrestricted bound [2, Theorem 1.1]. These are the computational dependencies of the present theorem. We also use Shevtsova's general-independent estimate

$$\Delta \leq C_E \ell + 0.2538 \ell^{4/3} \quad (\ell \leq 10^{-3}) \quad (1.4)$$

[7, Corollary 4.18, p. 303]. Its role is to localize a maximizing counterexample. Its positive remainder is removed by the subsequent support and local comparisons.

For the lattice step, define

$$p_E = \frac{4 - \sqrt{10}}{2} = \frac{1 - 1/c_E}{2}, \quad q_E = 1 - p_E, \quad \tau_E = p_E^2 + q_E^2, \quad h_E = 1/\tau_E. \quad (1.5)$$

Esseen's moment inequality [3, pp. 161–162] states that a centered variable  $Y$  supported on a translate of  $h\mathbb{Z}$  satisfies

$$|\mathbb{E}Y^3| + 3h\mathbb{E}Y^2 \leq c_E \mathbb{E}|Y|^3. \quad (1.6)$$

For positive variance, equality holds precisely for  $Y \stackrel{d}{=} \pm h(B - p_E)$ ,  $B \sim \text{Bern}(p_E)$ . Appendix B proves the quantitative form needed here. The iid asymptotic formula for this law gives a ratio tending to  $C_E$  and a Lyapunov ratio tending to zero [3, 5]. This proves optimality.

We use closed and strict CDFs  $F_S(x) = \mathbb{P}(S \leq x)$  and  $F_S^-(x) = \mathbb{P}(S < x)$ . For a centered sum of variance  $B^2$ ,

$$\sup_x |F_S(x) - \Phi(x/B)| = \sup_x \max\{F_S(x) - \Phi(x/B), \Phi(x/B) - F_S^-(x)\}. \quad (1.7)$$

Reflection interchanges these branches. The comparisons below retain these endpoint conventions and therefore include laws with atoms.

Section 2 selects an attained counterexample with small Lyapunov ratio and bounds every support. Section 3 finds an approximate period by a finite procedure. Section 4 converts its moment information into restrictions on every support point. Section 5 proves the independent local comparison with explicit parameters, and Section 6 evaluates their composition. The appendices give the Fourier estimates, the quantitative support argument, and the conditional-moment estimates used in these sections.

## 2 Selection and support equations

Normalize total variance to one, so  $\ell = T$ . For a fixed number of coordinates, maximize the positive signed ratio at a threshold, less a nondecreasing penalty on  $\ell$ .

**Lemma 2.1** (Localized selection). *Let  $0 < \eta \leq 1/100$  and  $0 < t \leq \min\{1/2000, \eta^3/108\}$ . If a finite array has  $\ell \leq t$  and  $\mathcal{R} > C_E$ , there is an attained penalized maximizer with the same number of coordinates, a threshold  $z$ , and*

$$C_E < R < 9/20, \quad \ell < 2t, \quad R \leq K := R + \ell \psi_t'(\ell) \leq R + \eta. \quad (2.1)$$

Here  $R$  is its full ratio. If a coordinate has variance  $s$  and third absolute moment  $b$ , then

$$\psi_t\left(\frac{\ell - b}{(1 - s)^{3/2}}\right) - \psi_t(\ell) \leq 2\eta s. \quad (2.2)$$

*Proof.* Put  $f(v) = 0$  for  $v \leq 0$ ,  $f(v) = v^2/(1+v)$  for  $v > 0$ , and  $\psi_t(u) = 2u^{1/3}f(\log(u/t))$ . This penalty is continuously differentiable, nonnegative, nondecreasing, and zero for  $u \leq t$ . After reflection if necessary, the supremum of

$$[F_S(z) - \Phi(z)]/\ell - \psi_t(\ell) \quad (2.3)$$

over centered variance-one arrays of the specified length is above  $C_E$ . If the objective exceeds  $r_0 > 0$ , then  $n^{-1/2} \leq \ell \leq r_0^{-1}$ . Bounded third moments give coordinatewise weak subsequences and uniform integrability of squares, preserving means and total variance. Chebyshev bounds the thresholds because their discrepancies are at least  $r_0/\sqrt{n}$ . Moving-threshold Portmanteau for the closed CDF gives the required upper semicontinuity at a limit. A strict loss of third moments would increase the positive ratio and decrease the penalty, contradicting the supremum. Thus the limit attains it. This is the fixed-dimension argument of He and Cheng [4, Lemma 4.1], applied coordinatewise.

Since  $f(\log 2) > 4/15 > 1/4$ , one has  $\psi_t(u) > u^{1/3}/2$  for  $u \geq 2t$ . For  $u \leq 10^{-3}$  this exceeds the excess in (1.4); for  $u \geq 10^{-3}$  it is at least  $1/20 > 9/20 - C_E$ . Hence the maximizer has  $\ell < 2t$ . For  $t < u \leq 4t$ ,  $f < 1$  and  $0 \leq f' \leq 1$  give

$$0 \leq u\psi'_t(u) \leq \frac{8}{3}(4t)^{1/3} < 3(4t)^{1/3} \leq \eta.$$

This holds also for  $u \leq t$ . Each coordinate has  $s^{3/2} \leq \ell < 10^{-3}$ , so  $s \leq 1/4$  and  $\ell_- = (\ell - b)/(1 - s)^{3/2} < (8/5)\ell < 4t$ . If  $\ell_- > \ell$ , integration on this actual interval gives

$$\psi_t(\ell_-) - \psi_t(\ell) \leq \eta \log(\ell_-/\ell) \leq \frac{3}{2}\eta[-\log(1-s)] \leq 2\eta s.$$

Otherwise monotonicity suffices. A larger discrepancy in the selected array, reflected for the other branch, would increase its objective; therefore  $R$  is its full ratio.  $\square$

Write  $s_i = \mathbb{E}X_i^2$ ,  $b_i = \mathbb{E}|X_i|^3$ ,  $M_i = \mathbb{E}[X_i|X_i|]$ , and let  $G_i$  be the CDF of  $S - X_i$ .

**Lemma 2.2** (Support contact). *For every  $y \in \mathbb{R}$ ,*

$$\begin{aligned} D_i(y) &:= G_i(z - y) - F_S(z) + \phi(z)y + \frac{z\phi(z)}{2}(y^2 - s_i) \\ &\quad + \frac{3}{2}K\ell(y^2 - s_i) - K\{|y|^3 - b_i - 3M_iy\} \leq 0. \end{aligned} \quad (2.4)$$

*Equality holds at every  $y \in \text{supp } X_i$ .*

*Proof.* Replace the raw law of coordinate  $i$  by  $(1 - e)\text{Law}(X_i) + e\delta_y$ , retain the raw threshold, and center and standardize the new array. The derivatives at  $e = 0$  of its mean, variance, and total third absolute central moment are  $y$ ,  $y^2 - s_i$ , and  $|y|^3 - b_i - 3M_iy$ . The last differentiation is justified by a bound  $C_y(1 + X_i^2)$ . The raw CDF derivative is  $G_i(z - y) - F_S(z)$ . Differentiating (2.3) and multiplying by  $\ell$  gives (2.4), with its sign supplied by maximality. Its integral under  $\text{Law}(X_i)$  is zero. Upper semicontinuity of  $D_i$  then extends equality to the entire support. This is the independent-coordinate form of the contamination argument of He and Cheng [4, Lemma 4.3].  $\square$

**Lemma 2.3** (Uniform support bound). *Every nonzero coordinate of the selected maximizer satisfies*

$$b_i < 3\ell s_i, \quad s_i < 9\ell^2, \quad \text{supp } X_i \subset (-8\ell, 8\ell). \quad (2.5)$$

*Proof.* Deletion, padded by a zero coordinate, stays in the maximization class. Its ratio at every threshold is at most  $R + 2\eta s_i$ . Compare the convolution of the deleted normal approximation with  $X_i$  to its convolution with  $N(0, s_i)$ . Put  $a = \phi(0)(1 + 4\phi(0))/6 < 13/75$ . Translation Taylor expansion and  $\mathbb{E}|N(0, s_i)|^3 = 4\phi(0)s_i^{3/2} \leq 4\phi(0)b_i$  give

$$R\ell(1 - s_i)^{3/2} \leq (R + 2\eta s_i)(\ell - b_i) + ab_i.$$

Using  $1 - (1 - s_i)^{3/2} \leq 3s_i/2$  and rearranging gives

$$(R - a)b_i \leq (3R/2 + 2\eta)\ell s_i.$$

Since  $R > 2/5$ , the right coefficient is less than  $3(R - a)$ . This proves  $b_i < 3\ell s_i$ , and Jensen gives  $\sqrt{s_i} < 3\ell$ .

For  $0 \leq s \leq 1/4$  put

$$H_s(z, y) = \Phi\left(\frac{z - y}{\sqrt{1 - s}}\right) - \Phi(z) + \phi(z)y + \frac{1}{2}z\phi(z)(y^2 - s).$$

Translation and scale Taylor expansions give

$$|H_s(z, y)| \leq \phi(0)|y|^3/6 + \phi(0)s|y|/2 + s^2/6, \quad (2.6)$$

$$|H_s(z, y) - \phi''(z)(sy/2 - y^3/6)| \leq y^4/36 + sy^2/6 + s^2/6. \quad (2.7)$$

Use  $\|\phi''\|_\infty = \phi(0)$ ,  $\|\phi'''\|_\infty \leq 2/\pi < 2/3$ ; the scale remainder is bounded in absolute value by  $4s^2/27 < s^2/6$ . Substitute the deleted bound in the contact equality. Its constant terms are

$$R(\ell_- - \ell) - \frac{3}{2}K\ell s + Kb + [\psi_t(\ell_-) - \psi_t(\ell)]\ell_-.$$

Since  $(1 - s)^{-3/2} < 8/5$  and  $(1 - s)^{-3/2} - 1 - 3s/2 \leq 6s^2$ , every support point satisfies

$$0 \leq H_s(z, y) - K|y|^3 + \frac{3}{2}K\ell y^2 + 3KM_i y + 6R\ell s^2 + \eta b + 4\eta\ell s. \quad (2.8)$$

The contact equation first bounds the support, since its negative cubic term dominates at infinity. Let  $A_i$  be its maximum absolute value. Use  $|M_i| \leq s$ ,  $b \leq sA_i$ ,  $K \leq 23/50$ , and (2.6) to obtain

$$A_i^3/3 \leq (7/10)\ell A_i^2 + (8/5)sA_i + (3/5)s^2 + (1/25)\ell s.$$

With  $s \leq 9\ell^2$ ,  $\ell \leq 1/8$ , and  $x = A_i/\ell$ ,

$$x^3/3 \leq 7x^2/10 + 72x/5 + 1287/200.$$

The difference is  $2539/600 > 0$  at  $x = 8$  and has positive derivative for  $x \geq 8$ . Thus  $A_i < 8\ell$ .  $\square$

### 3 An approximate period and uniform jitter

A uniform random shift suppresses the nonzero lattice peaks of a characteristic function. Its period must be chosen from the array itself. The following finite construction gives an explicit error.

**Lemma 3.1** (Finite period selection). *Let  $\mu$  be a probability measure on  $[-M, M]$ ,  $M > 0$ , and put*

$$g(u) = \int \frac{|e^{iud} - 1|^2}{d^2} \mu(dd), \quad q(u) = \sqrt{g(u)}, \quad \delta = 1/(4M),$$

where the integrand at zero is  $u^2$ . For  $T \geq \delta$  and  $0 < r \leq \delta/4$  define

$$B_0 = T/\delta + 2, \quad J = \lceil \log_2(T/\delta) \rceil + 1, \quad \varepsilon = r/(2B_0^J), \quad E = \{u \in [\delta, T] : q(u) \leq \varepsilon\}.$$

If  $E \neq \emptyset$ , there is  $p \in [\delta, T]$  such that  $q(p) \leq r/2$  and  $\text{dist}(u, p\mathbb{Z}) < r$  for every  $u \in E$ . For  $h = 2\pi/p$  and  $\ell > 0$ ,

$$\int_{\delta}^T e^{-g(u)/(2\ell^2)} |\text{sinc}(hu/2)| \frac{du}{u} \leq \frac{56\sqrt{\pi}}{9} \frac{r\ell}{p^2} + \log(T/\delta) e^{-\varepsilon^2/(2\ell^2)}. \quad (3.1)$$

If  $E = \emptyset$ , take  $h = 0$ ; the exponential term alone suffices.

*Proof.* The Hilbert-space triangle inequality gives subadditivity of  $q$ ,  $q(-u) = q(u)$ , and  $q(u) \leq |u|$ . Thus  $q$  is 1-Lipschitz. Also  $g'' \geq 2 - M^2g$  and  $g(u) \geq u^2/2$  for  $|u| \leq \delta$ . Start with  $p_0 \in E$ . If  $v \in E$  has distance at least  $r$  from  $p_j\mathbb{Z}$ , choose the nearest integer  $k$  and put  $p_{j+1} = |v - kp_j|$ . While  $p_j \geq \delta$ ,

$$r \leq p_{j+1} \leq p_j/2, \quad q(p_{j+1}) \leq \varepsilon + (T/\delta + 1/2)q(p_j) \leq B_0^{j+1}\varepsilon.$$

A proposed value in  $[r, \delta]$  would satisfy both  $q \leq r/2$  and  $q \geq r/\sqrt{2}$ . There are at most  $\lceil \log_2(T/\delta) \rceil$  halvings before this happens. The construction therefore stops with the required  $p$ .

Retain each  $I_k = [kp - r, kp + r] \cap [\delta, T]$ ,  $k \geq 1$ , that meets  $E$ . On it  $q \leq \varepsilon + 2r \leq 5\delta/8$ , so  $g'' \geq 1$ . At its minimum  $a_k$ , including an endpoint minimum,  $g(u) \geq (u - a_k)^2/2$ . Moreover  $|\text{sinc}(\pi u/p)| \leq r/u$  and  $u \geq (k - 1/4)p$ . Its integral is at most  $2\sqrt{\pi}\ell r/[(k - 1/4)^2 p^2]$ . Sum using  $\sum_{k \geq 1} (k - 1/4)^{-2} \leq 28/9$ . Outside the retained intervals  $g > \varepsilon^2$ , proving the assertion.  $\square$

**Proposition 3.2** (Uniform jitter). *Suppose  $\sum \mathbb{E}X_i^2 = 1$ ,  $\ell = \sum \mathbb{E}|X_i|^3 \leq 1/8$ , and  $|X_i| \leq 8\ell$ . For  $0 < \rho \leq 1$  put*

$$\begin{aligned} \delta &= 1/64, & T &= 80/\rho, & r &= \rho\delta^2/48, \\ B_0 &= T/\delta + 2, & J &= \lceil \log_2(T/\delta) \rceil + 1, & \varepsilon &= r/(2B_0^J), \\ L_{\text{jit}}(\rho) &= \min \left\{ \frac{1}{8}, \frac{\rho}{48000}, \frac{\rho\varepsilon^2}{8\log(T/\delta)} \right\}. \end{aligned} \quad (3.2)$$

If  $\ell \leq L_{\text{jit}}(\rho)$ , there is  $0 \leq h \leq 128\pi$  such that, for an independent  $U_h$  uniform on  $[-h/2, h/2]$ ,

$$\left\| F_{S+\ell U_h}(x) - \Phi(x) - \frac{\kappa}{6}(1-x^2)\phi(x) \right\|_{\infty} \leq \rho\ell, \quad \kappa = \sum_i \mathbb{E}X_i^3. \quad (3.3)$$

When  $h > 0$ ,  $g(2\pi/h) \leq r^2/4$  for the measure below. Here  $U_0 = 0$ .

*Proof.* Set  $Y_i = X_i/\ell$ , with independent copies  $Y'_i$ . The measure

$$\mu(A) = \frac{\ell^2}{2} \sum_i \mathbb{E}[(Y_i - Y'_i)^2 \mathbf{1}_{\{Y_i - Y'_i \in A\}}] \quad (3.4)$$

is a probability measure on  $[-16, 16]$ . It satisfies

$$g(u) = \ell^2 \sum_i (1 - |\mathbb{E}e^{iuY_i}|^2), \quad |\mathbb{E}e^{iuS/\ell}| \leq e^{-g(u)/(2\ell^2)}.$$

Apply Lemma 3.1. The low-frequency and signed-smoothing estimates in Appendix A bound the norm in (3.3) by

$$12000\ell^2 + 12r\ell/\delta^2 + \log(T/\delta)e^{-\varepsilon^2/(2\ell^2)} + 20\ell/T. \quad (3.5)$$

Each term is at most  $\rho\ell/4$  under (3.2); for the third use  $e^{-x} \leq 1/x$ .  $\square$

## 4 From moment stability to complete supports

The contact equations strengthen the moment estimates into bounds for each coordinate's full support.

**Proposition 4.1** (Two-cluster reduction). *Let  $0 < \lambda \leq 1$  and  $N \geq 1$ . Set*

$$\rho = 10^{-26}\lambda^2, \quad \eta = 10^{-10}\lambda, \quad L_* = \min\{L_{\text{jit}}(\rho), 1/(2\sqrt{N})\}. \quad (4.1)$$

*An attained maximizer of Lemma 2.1 with this penalty bound and  $\ell \leq L_*$  can be represented, after division by  $\ell$ , as independent coordinates*

$$Y_i = h_i(B_i - p_i) + U_i \quad (1 \leq i \leq m), \quad Z_1, \dots, Z_s, \quad (4.2)$$

where  $m \geq N$ ,  $h > 0$ ,  $B_i \sim \text{Bern}(p_i)$ , and

$$p_i \in [1/4, 3/4], \quad |h_i/h - 1| \leq \lambda, \quad \mathbb{E}(U_i | B_i) = 0, \quad |U_i| \leq \lambda h, \quad (4.3)$$

$$\mathbb{E}Z_j = 0, \quad |Z_j| \leq \lambda h, \quad \sum_j \text{Var } Z_j \leq \lambda m h^2. \quad (4.4)$$

*The representations are coordinatewise. Each pair  $(B_i, U_i)$  is independent of the other pairs and of all  $Z_j$ .*

The proof is in Appendix B. Rounding at the period from Proposition 3.2 loses a controlled total second moment. The deficit in Esseen's inequality forces that period close to  $h_E$  and the maximizing threshold close to zero. Using the same jitter for every deletion shows that its one-sided smoothing loss is small at each support point. Comparing two such losses restricts every difference of support points to a small neighborhood of  $h\mathbb{Z}$ . The cubic expansion of (2.8) bounds each scaled support diameter by  $7/2$  and puts the two-cluster probabilities in  $[1/4, 3/4]$ . Coordinates with small diameter have small total variance by moment stability. No lower bound on individual atom masses enters the argument.

## 5 An exact comparison near a common diameter

**Theorem 5.1** (Explicit independent local comparison). *Set*

$$\mathcal{A} = e^{10000}, \quad \lambda = e^{-\mathcal{A}}, \quad N = \lceil e^{\mathcal{A}} \rceil. \quad (5.1)$$

*Every independent array of the form (4.2)–(4.4), with  $m \geq N$ , has  $\mathcal{R} \leq C_E$ . Its ratio uses the total variance and the sum of third absolute moments of all the original coordinates.*

We follow the two residual regimes of He and Cheng [4, Section 3]. The estimates below are uniform in unequal probabilities, unequal gaps, and the additional coordinates.

### 5.1 Normalization and conditional estimates

Put  $v_i = p_i(1 - p_i)$  and define the variance-weighted mean gap

$$\bar{h} = \frac{\sum_i v_i h_i}{\sum_i v_i}.$$

Then  $(1 - \lambda)h \leq \bar{h} \leq (1 + \lambda)h$ . Divide every coordinate by  $\bar{h}$ , retaining  $h_i, U_i, Z_j$  for the rescaled gaps and residuals. Write

$$\begin{aligned} V &= \sum_i v_i, \quad \mu = \sum_i p_i, \quad h_i = 1 + a_i, \quad \sum_i v_i a_i = 0, \\ K_0 &= \sum_i B_i, \quad A = \sum_i a_i(B_i - p_i), \quad U = \sum_i U_i + \sum_j Z_j, \quad W = A + U, \quad S = K_0 + W, \\ \Lambda_A &= \sum_i v_i a_i^2, \quad \Lambda_U = \sum_i \mathbb{E}U_i^2 + \sum_j \mathbb{E}Z_j^2, \quad \Lambda = \Lambda_A + \Lambda_U, \quad B^2 = V + \Lambda. \end{aligned}$$

The uncentered sum  $S$  has mean  $\mu$ , and  $\text{Cov}(K_0, W) = 0$ . Put  $\varepsilon_0 = \max(\max_i \|U_i\|_\infty, \max_j \|Z_j\|_\infty)$ , with an empty maximum zero, and  $\xi = \Lambda/V$ . The hypotheses imply

$$3m/16 \leq V \leq m/4, \quad \max_i |a_i| \leq 3\lambda, \quad \varepsilon_0 \leq 2\lambda, \quad \xi \leq 8\lambda, \quad \Lambda \leq 2\lambda m. \quad (5.2)$$

For example the upper bound before enlargement is  $\xi \leq [\lambda + (7/4)\lambda^2]/[(3/16)(1 - \lambda)^2]$ .

Let  $T_X$  and  $\kappa$  be the total absolute and signed third central moments of the original array. Its reference Bernoulli moments are

$$T_0 = \sum_i v_i(p_i^2 + (1 - p_i)^2), \quad \kappa_0 = \sum_i v_i(1 - 2p_i).$$

Conditional centering gives

$$|T_X - T_0|, \quad |\kappa - \kappa_0| \leq 2\sqrt{m\Lambda_A} + 4\Lambda, \quad (5.3)$$

$$T_X \geq V/3, \quad 1/4 \leq T_X/B^2 \leq 2, \quad B^2 \leq m, \quad B/2 \leq P := B^3/T_X \leq 4B. \quad (5.4)$$

Indeed  $|(1 + a)^3 - 1| \leq 4|a|$  and  $\sum v_i |a_i| \leq \sqrt{V\Lambda_A}$ . Within each cluster the sign is fixed: its unperturbed magnitude is at least  $7/32 > \varepsilon_0$ . Expanding the cubic changes either third moment by at most  $4\mathbb{E}U_i^2$ , since the linear term vanishes. An extra coordinate contributes at most  $\varepsilon_0 \mathbb{E}Z_j^2$ . Jensen gives  $T_X \geq (7/8)^3 T_0 \geq 343V/1024 > V/3$ . Every original coordinate has absolute value at most two, so  $T_X \leq 2B^2$ , proving (5.4).

For  $m \geq 10^{12}$  and  $|k - \mu| \leq 6\sqrt{m}$ , Appendix C proves

$$|m_k| := |\mathbb{E}(A \mid K_0 = k)| \leq 10^4 \sqrt{\Lambda_A}/\sqrt{m}, \quad (5.5)$$

$$\begin{aligned} |\text{Var}(A \mid K_0 = k) - \Lambda_A| &\leq 40000\Lambda_A/\sqrt{m}, \\ \mathbb{E}[(A - m_k)^4 \mid K_0 = k] &\leq 10^7 \Lambda_A^2. \end{aligned} \quad (5.6)$$

It also proves the relative label bound

$$\left| \frac{\mathbb{P}(B_i = b \mid K_0 = k)}{\mathbb{P}(B_i = b)} - 1 \right| \leq 500/\sqrt{m} \quad (b = 0, 1). \quad (5.7)$$

For  $\Lambda > 0$  define

$$D = \Lambda^2 + \varepsilon_0^2 \Lambda_U, \quad H = \max \left\{ \sqrt{\Lambda_A}, \frac{\Lambda}{\sqrt{\Lambda + \varepsilon_0^2}} \right\}. \quad (5.8)$$

Then, in the same count range,

$$\begin{aligned} \mathbb{E}(W \mid K_0 = k) &= m_k, \quad \Lambda/2 \leq \text{Var}(W \mid K_0 = k) \leq 2\Lambda, \\ \mathbb{E}[(W - m_k)^4 \mid K_0 = k] &\leq 10^8 D, \quad \mathbb{E}(|W - m_k| \mid K_0 = k) \geq H/30000, \\ \mathbb{E}(W^4 \mid K_0 = k) &\leq 10^9 D, \quad \mathbb{E}W^4 \leq 400D. \end{aligned} \quad (5.9)$$

To see the extension, condition first on all labels. The noise then has variance  $T_{\mathbf{b}} \leq 4\Lambda_U$  and fourth moment at most  $3T_{\mathbf{b}}^2 + \varepsilon_0^2 T_{\mathbf{b}}$ . Its mean given  $K_0$  is zero, and (5.7) compares its variance to  $\Lambda_U$ . Combine these facts with (5.6) and  $|x+y|^4 \leq 8(|x|^4 + |y|^4)$ . Interpolation  $\mathbb{E}|Y| \geq (\mathbb{E}Y^2)^{3/2}/(\mathbb{E}Y^4)^{1/2}$  supplies the second term in  $H$ . Conditional Jensen given the labels supplies the first from  $A - m_k$ . In particular  $|m_k| \leq 10^4 H/\sqrt{m}$  uniformly, even when  $\Lambda_A$  is much smaller than  $\varepsilon_0^2/m$ . Finally  $\max a_i^2 \leq 16\Lambda_A/3$  gives  $\mathbb{E}A^4 \leq 9\Lambda_A^2$ , and hence the unconditional bound.

## 5.2 Uniform approximation and central thresholds

Let  $J$  be the CDF of  $S$  plus an independent uniform variable on  $[-1/2, 1/2]$ . If  $T \geq 10$  and  $\xi(T+1)^2 \leq 1/56$ , put

$$E_m(T) = \frac{401(1 + \log T)}{\sqrt{m}} + (1 + \log T)\xi(T+1)^2 + 60000(1 + \log T)\sqrt{m}e^{-m/262144} + 56/T. \quad (5.10)$$

Appendix A proves, for  $z = (x - \mu)/B$ ,

$$\sqrt{m} \left\| J(x) - \Phi(z) - \frac{\kappa}{6B^3}(1 - z^2)\phi(z) \right\|_{\infty} \leq E_m(T). \quad (5.11)$$

The expansion includes the third moments of the additional coordinates.

Fix

$$a_0 = e^{1000}, \quad \mathcal{C} = e^{1004}, \quad \mathcal{B} = e^{1006}, \quad q = e^{-\mathcal{C}}, \quad T = e^{\mathcal{B}}.$$

The inequalities  $\mathcal{A} > 100\mathcal{B}$ ,  $\mathcal{B} > 7\mathcal{C}$ , and  $\mathcal{C} > 50a_0$  imply, for every  $m \geq N$ ,

$$\xi(T+1)^2 < 1/56, \quad E_m(T) < q/1000, \quad m^{-1/2} < q/1000, \quad 200\lambda < q/1000. \quad (5.12)$$

For a direct check, the four terms in (5.10) are at most

$$802\mathcal{B}e^{-\mathcal{A}/2}, \quad 64\mathcal{B}e^{-\mathcal{A}+2\mathcal{B}}, \quad 240000\mathcal{B}(262144)^2e^{-3\mathcal{A}/2}, \quad 56e^{-\mathcal{B}}.$$

Each is below  $q/4000$ . The third uses  $e^{-x} \leq 2/x^2$ , so this verification places no upper bound on  $m$ .

Shift the jitter by  $\pm 1/2$ . Either normalized signed discrepancy is at most

$$\phi(z) \left\{ \frac{B^2}{2T_X} + \frac{|\kappa|}{6T_X}|1 - z^2| \right\} + 4E_m(T) + m^{-1/2}. \quad (5.13)$$

The normalized Taylor error is at most  $1/(4B) < m^{-1/2}$ , using  $\|\phi'\|_{\infty} < 1/4$ ,  $\|((1 - z^2)\phi(z))'\|_{\infty} < 5/4$ , and (5.4). The main term is at most  $(3 + z^2)\phi(z)$ , below  $1/100$  for  $|z| \geq 4$ . Thus the full bound there is below  $1/24$ .

## 5.3 Small residual variance

Suppose  $0 < \Lambda \leq e^{-1000}$ . If  $\Lambda = 0$ , the result is exactly (1.2). We use a one-sided loss estimate: if  $\mathbb{E}Y = 0$ ,  $\mathbb{E}Y^4 \leq 1/160000$ ,  $f(0) = 0$ , and  $3/4 \leq f' \leq 5/4$  on  $[-7/10, 7/10]$ , then for  $|u| \leq 3/5$ ,

$$\begin{aligned} \mathbb{P}(Y > u) + f(u) &\geq \frac{3}{8}\mathbb{E}|Y| - 176000\mathbb{E}Y^4, \\ \mathbb{P}(Y < u) - f(u) &\geq \frac{3}{8}\mathbb{E}|Y| - 176000\mathbb{E}Y^4. \end{aligned} \quad (5.14)$$

This is the clipping argument of He and Cheng [4, Lemma 3.2 and equation (83)]. Appendix C gives its proof with these constants and the stated threshold range.

The count estimate in Appendix A gives

$$b_k := \mathbb{P}(K_0 = k) \geq e^{-100}/\sqrt{m} \quad (|k - \mu| \leq 6\sqrt{m}). \quad (5.15)$$

Write a threshold with  $|z| < 4$  as  $k + u$ ,  $|u| \leq 1/2$ , where  $k$  is a nearest integer. Then  $|k - \mu| \leq 5\sqrt{m}$ . For  $f_k(u) = [\Phi((k - \mu + u)/B) - \Phi((k - \mu)/B)]/b_k$ , we have  $3/4 \leq f'_k \leq 5/4$  on  $[-1, 1]$ . Indeed the relative count error is at most  $18e^{100}/\sqrt{m}$ , and the logarithm of the Gaussian density ratio is at most  $100\xi + 30/\sqrt{m}$ ; both are below  $1/100$ .

The exact count decompositions are

$$\begin{aligned} F_S(k + u) &= F_{K_0}(k) - b_k \mathbb{P}(W > u \mid K_0 = k) + e_{k,u}^+, \\ F_S^-(k + u) &= F_{K_0}(k - 1) + b_k \mathbb{P}(W < u \mid K_0 = k) + e_{k,u}^-. \end{aligned} \quad (5.16)$$

Each remainder is bounded in absolute value by

$$\sum_{j \neq k} b_j \mathbb{P}\{|W| \geq |j - k| - 1/2 \mid K_0 = j\} \leq 2 \cdot 10^{11} D/\sqrt{m}.$$

For  $|j - k| \leq \sqrt{m}$ , use  $b_j \leq 3/\sqrt{m}$ , (5.9), and  $\sum_{r \neq 0} (|r| - 1/2)^{-4} < 40$ . For more distant counts use  $\mathbb{E}W^4 \leq 400D$  and a denominator at least  $m^2/16$ .

Apply (5.14) to  $Y = W - m_k$  conditionally on  $K_0 = k$ , using  $f_k(v + m_k) - f_k(m_k)$  and  $v = u - m_k$ . The bounds above give  $|m_k| < 1/20$  and the required fourth-moment bound. Also  $e^{-100}/5 \leq Pb_k \leq 5$ . The resulting normalized loss is at least  $e^{-120}H$ . Its fourth-moment and leakage errors are below  $10^{14}D$ , and its conditional-mean error is at most  $62500\sqrt{\Lambda_A}/\sqrt{m}$ .

Let  $R_0 = \mathcal{R}(B_1 - p_1, \dots, B_m - p_m) \leq C_E$ . Equation (5.3) gives

$$\left| \frac{B^3/T_X}{V^{3/2}/T_0} - 1 \right| \leq 32\sqrt{\Lambda_A}/\sqrt{m} + 96\Lambda/m.$$

The normal variance change costs at most  $\xi/8$  before multiplication by  $P$ . Combining these errors with (5.16), either central normalized branch is at most

$$R_0 - e^{-120}H + e^{40} \left\{ D + \frac{\sqrt{\Lambda_A} + \Lambda}{\sqrt{m}} \right\}. \quad (5.17)$$

The full definition of  $H$  gives

$$D/H \leq (\Lambda + \varepsilon_0^2)^{3/2}, \quad \sqrt{\Lambda_A}/H \leq 1, \quad \Lambda/H \leq \sqrt{\Lambda + \varepsilon_0^2}.$$

Since  $\varepsilon_0 \leq e^{-1000}$  and  $m \geq e^{1000}$ , the error is below  $e^{40}(3e^{-1500} + 2e^{-500})H < e^{-120}H/2$ . Thus the central comparison is strict for every positive residual variance. Finally  $R_0 \geq \phi(0)/4 > 1/12$ : apply the count estimate at an integer nearest  $\mu$  and take half its jump. The bound  $1/24$  at distant thresholds is also below  $R_0$ .

## 5.4 Positive residual variance and interval mass

Suppose  $\Lambda \geq e^{-1000}$  and  $|z| \leq 4$ . Choose all integers  $|k - x| \leq \max(1, \sqrt{\Lambda})$ . There are at least  $\sqrt{\Lambda}$  of them, all within  $6\sqrt{m}$  of  $\mu$ . Tilt the Bernoulli probabilities exponentially to make their count

mean  $k$ , and project the gap residual off the count, as in Appendix C. On  $K_0 = k$  the residual is  $m_k^0 + R_k$ , where under the tilted measure  $R_k$  is a sum of independent centered pair residuals and additional coordinates, with

$$L := \text{Var}_\theta R_k \in [\Lambda/2, 2\Lambda], \quad \text{Var}_\theta K_0 \geq m/10, \quad |m_k^0| \leq \sqrt{\Lambda}, \quad \text{Cov}_\theta(K_0, R_k) = 0. \quad (5.18)$$

Each residual summand is bounded by  $8\lambda$ . The within-cluster noise variances change relatively by at most  $120/\sqrt{m}$  under the tilt; the extra coordinates do not change. Thus these assertions include their actual variances.

Apply Lemmas A.2 and A.3 to both quarter-intervals  $(1/4, 1/2)$  and  $(-1/2, -1/4)$ , with

$$L_0 = \frac{1}{2}e^{-1000}, \quad C_0 = 4 + 2e^{500}, \quad A_g = 4e^{500}, \quad \omega = 16e^{A_g^2/6}, \quad T_g = 4\omega.$$

The shift  $y = x - k - m_k^0$  satisfies  $|y| \leq C_0\sqrt{L}$ , and  $A_g \geq C_0 + (7/16)/\sqrt{L_0}$ . The interval minorant has Gaussian expectation at least  $\phi(A_g)/(4\omega\sqrt{L})$ . With  $\beta = 7/64$ , the three terms in (A.5) are at most

$$128e^{-\mathcal{A}/2}, \quad 128e^{500-\mathcal{A}}, \quad 10^7 e^{4a_0-\mathcal{A}}.$$

Their sum is below  $e^{-10a_0} < \phi(A_g)/32$ , because  $\phi(A_g) > e^{-9a_0}$ ,  $\omega < e^{3a_0}$ , and  $T_g < e^{4a_0}$ . Also  $8\lambda T_g < \beta/162$ . For the third term use  $L \leq m$ ,  $e^{-x} \leq 2/x^2$ , and  $m \geq N$ ; this is uniform over all possible residual variances.

Each conditional quarter-interval probability is at least  $\phi(A_g)/(8\omega\sqrt{L})$ . Multiply by the count probabilities (5.15) and sum over the chosen integers. Since  $L \leq 2\Lambda$  and  $\mathcal{C} > 50a_0$ , this gives

$$\mathbb{P}\{S - x \in (1/4, 1/2)\}, \quad \mathbb{P}\{S - x \in (-1/2, -1/4)\} \geq q/\sqrt{m}. \quad (5.19)$$

The two actual one-sided jitter losses therefore satisfy

$$J(x + 1/2) - F_S(x) \geq q/(2\sqrt{m}), \quad F_S^-(x) - J(x - 1/2) \geq q/(2\sqrt{m}).$$

After multiplication by  $P \geq B/2$ , each loss exceeds  $q/10$ . Esseen's Bernoulli inequality gives  $3V + |\kappa_0| \leq c_E T_0$ . Using (5.3), the main term of (5.13) is consequently at most

$$C_E + 20\sqrt{\Lambda_A/m} + 42\Lambda/m \leq C_E + 200\lambda.$$

Either central branch is below  $C_E - q/10 + 200\lambda + 4E_m(T) + m^{-1/2} < C_E - q/20$ . The distant thresholds were already bounded by  $1/24$ . This completes the proof of Theorem 5.1.

## 6 Evaluation of the threshold

Take  $\lambda, N$  from (5.1) and  $\rho, \eta, L_*$  from (4.1). A violation with Lyapunov ratio at most

$$t_* := \min\{L_*/2, \eta^3/108, 1/2000\} \quad (6.1)$$

would give, by Lemma 2.1, an attained violating maximizer with  $\ell < 2t_* \leq L_*$ . Proposition 4.1 puts the entire array into the class of Theorem 5.1, contradicting  $R > C_E$ .

It remains to show  $\ell_0 < t_*$ . Write  $A = \mathcal{A} = e^{10000}$ , so  $\lambda = e^{-A}$ ,  $N = \lceil e^A \rceil$ ,  $\rho = 10^{-26}e^{-2A}$ , and  $\eta = 10^{-10}e^{-A}$ . For the jitter parameters put

$$Q = T/\delta = 5120/\rho, \quad B_0 = Q + 2, \quad J = \lceil \log_2 Q \rceil + 1, \quad r = \rho/(48 \cdot 4096), \quad \varepsilon = r/(2B_0^J).$$

Since  $A > 1000$ ,  $\log 2 > 2/3$ , and  $\log 10 < 3$ ,

$$\log(1/\rho), \log Q, \log B_0, \log(1/r) < 3A, \quad J < 5A.$$

The ceiling is included:  $J < \log Q / \log 2 + 2 < 5A$ . Thus  $\log(1/\varepsilon) < 3A + 1 + 15A^2 < 16A^2$ . The negative logarithm of the third entry defining  $L_{\text{jit}}$  is

$$\log(1/\rho) + 2\log(1/\varepsilon) + \log(8\log Q) < 32A^2 + 4A < 40A^2.$$

The first two entries have smaller negative logarithms. Also  $N < 2e^A$ , so  $\log(2\sqrt{N}) < A$  and  $-\log L_* < 40A^2$ . The other entries of (6.1) have negative logarithms

$$3A + 30\log 10 + \log 108 < 4A, \quad \log 2000 < A.$$

Therefore  $-\log t_* < 41A^2$ . Finally  $e^4 > \sum_{j=0}^5 4^j/j! = 643/15 > 41$ , whence

$$-\log \ell_0 = e^{20004} = e^4 A^2 > 41A^2 > -\log t_*.$$

This completes the proof of Theorem 1.1.

## A Fourier estimates with numerical constants

We use  $\hat{\nu}(t) = \int e^{itx} \nu(dx)$  and  $\text{sinc } u = \sin(u)/u$ , with its continuous value at zero. This appendix supplies the quantitative versions of the signed smoothing and jitter arguments used in Sections 3 and 5.

### A.1 Signed smoothing and the low-frequency expansion

If  $F$  is a probability CDF with finite first absolute moment and  $G$  is the CDF primitive of a finite signed measure of mass one and finite first absolute moment, with Lipschitz constant  $M_G$ , then

$$\|F - G\|_\infty \leq \frac{1}{4} \int_{-A}^A \frac{|\hat{F}(t) - \hat{G}(t)|}{|t|} dt + 24M_G/A. \quad (\text{A.1})$$

Here  $\hat{F}$  and  $\hat{G}$  denote the transforms of their measures. This is the signed smoothing lemma of He and Cheng [4, Lemma 2.1 and equation (71)]. Its proof uses the even density  $k(x) = 3\text{sinc}(x/4)^4/(8\pi)$ , whose transform is supported on  $[-1, 1]$ . The bound  $k(x) \leq 3\min(1, 256/x^4)/(8\pi)$  gives  $\mathbb{E}|Z| \leq 12/\pi < 4$ . With shift  $a = 16$ , its one-sided tail  $\varepsilon_k$  is below  $1/8$ , and  $C_a = \mathbb{E}(16 - Z)_+ < 18$ . At a positive near-maximum of  $F - G$ , monotonicity of  $F$  gives  $(F - G)(x + y) \geq (F - G)(x) - M_G y$  for  $y \geq 0$ . Average with the shifted kernel; use  $-\|F - G\|_\infty$  on its opposite tail. The negative branch uses the reflected shift. This bounds the norm by

$$\frac{\|(F - G) * k_A\|_\infty}{1 - 2\varepsilon_k} + \frac{C_a M_G}{A(1 - 2\varepsilon_k)}.$$

Fourier inversion of the integrable CDF difference gives coefficients  $1/[2\pi(1 - 2\varepsilon_k)] \leq 1/4$  and  $C_a/(1 - 2\varepsilon_k) \leq 24$ . Only  $F$  needed monotonicity. In particular the weaker coefficients  $1/\pi$  and 40 are valid.

Consider the variance-one row of Proposition 3.2. Write  $v_i = \mathbb{E}X_i^2$ ,  $\kappa = \sum \mathbb{E}X_i^3$ , and  $f_i(t) = \mathbb{E}e^{itX_i}$ . For  $|t| \leq 1/(64\ell)$ ,

$$|f_i(t) - 1| \leq v_i t^2/2 \leq 1/128, \quad \sum_i \mathbb{E}X_i^4 \leq 8\ell^2, \quad \sum_i v_i^2 \leq 64\ell^2.$$

Using  $|\log(1+z) - z| \leq |z|^2$  for  $|z| \leq 1/2$  yields

$$\sum_i \log f_i(t) = -t^2/2 + \kappa(it)^3/6 + w(t), \quad |w(t)| \leq (49/3)\ell^2 t^4 \leq t^2/6.$$

Thus the error against  $\hat{\nu}(t) = e^{-t^2/2}(1 + \kappa(it)^3/6)$  is at most

$$\ell^2 e^{-t^2/3} \{(49/3)t^4 + t^6/72\}. \quad (\text{A.2})$$

Multiplication by  $\text{sinc}(h\ell t/2)$  adds at most  $(h^2\ell^2/24)(t^2 + \ell|t|^5/6)e^{-t^2/2}$ . The positive-frequency integrals after division by  $t$  are at most  $(591/8)\ell^2$  and  $(13h^2/288)\ell^2$ , respectively. For  $h \leq 128\pi$  their sum is below  $7500\ell^2$ . The Gaussian comparison tail starting at  $a = 1/(64\ell)$  is at most

$$\ell^2 \{64^2 + 64/3\} < 4118\ell^2.$$

Here use  $\int_a^\infty e^{-t^2/2} dt/t \leq a^{-2}$  and  $\int_a^\infty t^2 e^{-t^2/2} dt \leq 2/a$ . The comparison CDF  $\Phi(x) + \kappa(1 - x^2)\phi(x)/6$  has Lipschitz constant below  $1/2$ , since  $\|\phi'''\|_\infty \leq 2/\pi$  and  $|\kappa| \leq 1/8$ . Applying the weaker smoothing coefficients  $1/\pi, 40$ , and Lemma 3.1, now proves (3.5).

For the local calculation a smaller low-frequency bound is convenient. Put  $m_{4,i} = \mathbb{E}X_i^4$  and  $A_i = -v_i t^2/2 + \mathbb{E}X_i^3(it)^3/6$ . On the same interval,  $|A_i| \leq (25/48)v_i t^2 < 1/64$  and

$$|f_i(t) - e^{A_i}| \leq m_{4,i} t^4/2.$$

Symmetrization gives  $|f_i(t)| \leq e^{-v_i t^2/3}$ . Telescope the products; omitting an individual decay factor costs less than two. Since  $\sum_i m_{4,i} \leq 8\ell^2$ ,

$$|\hat{F}(t) - \hat{\nu}(t)| \leq 8\ell^2 t^4 e^{-t^2/3} + (\ell^2 t^6/72)e^{-t^2/2}.$$

Integrating over both signs gives less than  $73\ell^2$ . Using  $|\text{sinc}(h\ell t/2) - 1| \leq h^2\ell^2 t^2/24$  and the actual product bound adds at most  $h^2\ell^2/8$ . Hence

$$\int_{|t| \leq 1/(64\ell)} \frac{|\text{sinc}(h\ell t/2)\hat{F}(t) - \hat{\nu}(t)|}{|t|} dt \leq (73 + h^2/8)\ell^2. \quad (\text{A.3})$$

## A.2 The count estimate

**Lemma A.1.** *For a sum  $K$  of independent Bernoulli variables, put  $\mu = \mathbb{E}K$  and  $V = \text{Var } K$ . If  $V \geq 1/2$ , then*

$$\left| \mathbb{P}(K = k) - \frac{\phi((k - \mu)/\sqrt{V})}{\sqrt{V}} \right| \leq 6/V, \quad \sup_k \mathbb{P}(K = k) < 1/\sqrt{V}. \quad (\text{A.4})$$

*If the probabilities lie in  $[\alpha, 1 - \alpha]$ , let  $\beta = \alpha(1 - \alpha)$  and  $\gamma = \phi(M/\sqrt{\beta})$ . For  $m \geq 144/(\beta\gamma^2)$  and  $|k - \mu| \leq M\sqrt{m}$ ,  $\mathbb{P}(K = k) \geq \gamma/\sqrt{m}$ .*

*Proof.* For  $Z_i = B_i - p_i$  and  $v_i = p_i(1 - p_i)$ , on  $[-\pi, \pi]$ ,

$$|\mathbb{E}e^{itZ_i}| \leq e^{-2v_i \sin^2(t/2)} \leq e^{-2v_i t^2/\pi^2}, \quad |\mathbb{E}e^{itZ_i} - e^{-v_i t^2/2}| \leq v_i(|t|^3/6 + t^4/32).$$

Telescope, retaining the decay from  $V - v_i \geq V/2$ . The difference is at most  $V(1/6 + \pi/32)|t|^3 e^{-Vt^2/\pi^2}$ . Its integral divided by  $2\pi$ , plus the Gaussian tail, is at most  $(\pi^3/12 + \pi^4/64 + 1/\pi^2)/V < 6/V$ . Integrating the product modulus gives  $\sqrt{\pi}/(2\sqrt{2V}) < 1/\sqrt{V}$ . For the last assertion the error is at most half the central Gaussian term, and  $V \leq m/4$  supplies the stated factor.  $\square$

### A.3 Conditioning an independent product on a count

**Lemma A.2.** *Let  $(B_i, R_i)$  be independent pairs, with  $B_i \sim \text{Bern}(r_i)$ ,  $r_i \in [a, 1-a]$ ,  $\mathbb{E}R_i = 0$ ,  $|R_i| \leq d_0$ . Allow additional independent centered  $Z_j$  with  $|Z_j| \leq d_0$ . Put*

$$K = \sum_i B_i, \quad R = \sum_i R_i + \sum_j Z_j, \quad \mathbb{E}K = k \in \mathbb{Z}, \quad V = \text{Var } K \geq 1000, \quad L = \text{Var } R > 0,$$

and assume  $\text{Cov}(K, R) = 0$ . Set  $\beta = a(1-a)$ . If  $g$  is continuous and integrable, with integrable Fourier transform supported on  $[-T, T]$ , and  $d_0T \leq \beta/162$ , then

$$\begin{aligned} & \sup_y |\mathbb{E}[g(R-y) \mid K=k] - \mathbb{E}g(\sqrt{L}Z - y)| \\ & \leq \frac{\|g\|_1}{\sqrt{L}} \left\{ \frac{32}{\sqrt{V}} + \frac{8d_0}{\sqrt{L}} + 2T\sqrt{VL}e^{-\beta m/162} \right\}, \quad Z \sim N(0, 1). \end{aligned} \quad (\text{A.5})$$

*Proof.* Write  $\chi(s, t) = \mathbb{E}e^{is(K-k)+itR}$ . For  $|s| \leq 1/4$ ,  $|t| \leq T$ , each centered individual phase  $X$  is bounded by  $1/2$ . If  $v = \mathbb{E}X^2$  and  $b = \mathbb{E}|X|^3$ , symmetrization gives  $|\mathbb{E}e^{iX}| \leq e^{-v/3}$ , and  $v^2 \leq b/2$  gives  $|\mathbb{E}e^{iX} - e^{-v/2}| < b/4$ . The total phase variance is exactly  $Vs^2 + Lt^2$ , by the zero covariance. The third absolute moments sum to at most  $4(V|s|^3 + d_0L|t|^3)$ . Telescoping therefore gives

$$|\chi(s, t) - e^{-Vs^2/2 - Lt^2/2}| \leq 2(V|s|^3 + d_0L|t|^3)e^{-(Vs^2 + Lt^2)/3}.$$

Its double integral is at most  $56(VL)^{-1/2}(V^{-1/2} + d_0L^{-1/2})$ ; use  $\int e^{-u^2/3} du = \sqrt{3\pi}$ ,  $\int |u|^3 e^{-u^2/3} du = 9$ . For  $1/4 \leq |s| \leq \pi$ , a Bernoulli factor is at most  $e^{-2\beta/81} \leq 1 - \beta/81$  because  $\sin(1/8) > 1/9$ . The residual changes it by at most  $Td_0 \leq \beta/162$ , giving  $|\chi(s, t)| \leq e^{-\beta m/162}$ . Its double integral is at most  $4\pi T e^{-\beta m/162}$ . The omitted Gaussian count tail has integral at most  $8\sqrt{2\pi}(V\sqrt{L})^{-1}e^{-V/32}$ .

Fourier inversion in the count and residual uses  $|\hat{g}| \leq \|g\|_1$ . By Lemma A.1,  $q = \mathbb{P}(K = k)$  differs from  $q_0 = (2\pi V)^{-1/2}$  by at most  $6/V$ ;  $V \geq 1000 > 288\pi$  ensures  $q \geq q_0/2$ . Division by  $q$  bounds the three preceding errors by

$$\frac{8\|g\|_1}{\sqrt{L}}(V^{-1/2} + d_0L^{-1/2}), \quad 2\|g\|_1 T\sqrt{V}e^{-\beta m/162}, \quad \frac{3\|g\|_1}{\sqrt{VL}}e^{-V/32}.$$

Replacing  $q_0/q$  by one in the Gaussian expectation costs at most  $12\|g\|_1/\sqrt{VL}$ . Enlarge  $8 + 3 + 12$  to 32. The factors involving  $y$  have modulus one, proving uniformity.  $\square$

**Lemma A.3** (An interval minorant). *For an interval  $(a, b)$  put  $w_0 = (a+b)/2$ ,  $d = (b-a)/2$ . Fix  $L_0 > 0$ ,  $C_0 \geq 0$ , and choose*

$$A_g \geq C_0 + (|w_0| + d/2)/\sqrt{L_0}, \quad \omega = (2/d)e^{A_g^2/6}, \quad T = 4\omega.$$

Then

$$g(w) = \left(1 - \frac{(w - w_0)^2}{d^2}\right) \text{sinc}(\omega(w - w_0))^4$$

is at most  $\mathbf{1}_{(a,b)}$ , has integrable Fourier transform supported on  $[-T, T]$ , and satisfies

$$\|g\|_1 \leq 4/\omega, \quad \mathbb{E}g(\sqrt{L}Z - y) \geq \frac{\phi(A_g)}{4\omega\sqrt{L}} \quad (L \geq L_0, \quad |y| \leq C_0\sqrt{L}).$$

If the bracket in (A.5) is at most  $\phi(A_g)/32$ , the corresponding conditional interval probability is at least  $\phi(A_g)/(8\omega\sqrt{L})$ .

*Proof.* On  $|w - w_0| \leq 1/\omega$ , the quadratic factor is at least  $3/4$ ,  $\text{sinc} \geq 5/6$ , and hence  $g \geq 1/4$ . Its Gaussian density there is at least  $\phi(A_g)/\sqrt{L}$ . The negative part has integral at most  $2/(d^3\omega^4)$ , with density at most  $\phi(0)/\sqrt{L}$ . Since  $\omega^3 = 8\phi(0)/(d^3\phi(A_g))$ , subtraction leaves the asserted lower bound. The integrals of  $\min(1, |\omega s|^{-4})$  and  $s^2 \min(1, |\omega s|^{-4})$  are  $8/(3\omega)$  and  $8/(3\omega^3)$ ;  $\omega d \geq 2$  proves the  $L^1$  bound. The transform of the fourth power of  $\text{sinc}$  is a compactly supported cubic spline. Multiplication by the quadratic differentiates it twice, leaving an integrable transform on the same support. The final assertion follows from Lemma A.2.  $\square$

#### A.4 Jitter on the actual count lattice

He and Cheng [4, Lemma A.4] prove the iid jitter estimate. We use the cancellation  $\sum_i v_i a_i = 0$  to control unequal gaps. Use the normalization of Section 5, and let  $f_i$  include all original coordinate characteristic functions. Put

$$g(t) = \sum_i (1 - |f_i(t)|^2), \quad g_0(t) = 2V(1 - \cos t).$$

For  $\max |a_i|, \varepsilon_0 \leq 1/4$ ,

$$\begin{aligned} |g - g_0| &\leq \Lambda t^2, \\ |g' - g'_0| &\leq \Lambda(2|t| + 2t^2), \\ |g'' - g''_0| &\leq \Lambda(2 + 8|t| + 4t^2). \end{aligned} \tag{A.6}$$

To prove these bounds, couple each actual symmetrized difference with  $D_0 = B_i - B'_i$ , writing it as  $D_0 + E$ , where  $E = a_i D_0 + U_i - U'_i$ . For an extra coordinate use  $D_0 = 0$ . All interpolation points lie in  $[-2, 2]$ , and  $\sum \mathbb{E} E^2 = 2\Lambda$ . Taylor-expand  $1 - \cos(td)$ ,  $d \sin(td)$ , and  $d^2 \cos(td)$  in  $d$ . The terms involving  $U$  vanish conditionally to first order, and the remaining first-order terms cancel by  $\sum_i v_i a_i = 0$ . The second derivatives give exactly (A.6).

If  $T \geq 10$  and  $\xi(T+1)^2 \leq 1/56$ , each interval  $I_j = [2\pi j - 1/4, 2\pi j + 1/4]$  meeting  $[-T, T]$  has a unique minimum  $t_j$  of  $g$ , with

$$|t_j - 2\pi j| \leq 4\Lambda(T+1)^2/V, \quad \left| \prod_i f_i(t) \right| \leq e^{-V(t-t_j)^2/4} \quad (t \in I_j).$$

Indeed  $g'' \geq 27V/16 \geq V$  there, and the endpoint derivatives have opposite signs: their reference magnitudes exceed  $V/3$  while the errors are at most  $V/14$ . Outside these intervals  $g \geq V/25$ , giving product modulus at most  $e^{-V/50}$ . For  $j \neq 0$ , the raw jitter multiplier  $\text{sinc}(t/2)$  vanishes at  $2\pi j$  and has Lipschitz constant at most  $1/4$ . Using  $|t| \geq 5|j|$ , integrate its linear bound against the displayed Gaussian and sum the harmonic series. The result over both signs is

$$\sum_{j \neq 0} \int_{I_j \cap [-T, T]} \frac{|\text{sinc}(t/2) \prod_i f_i(t)|}{|t|} dt \leq (1 + \log T) \left\{ \frac{2}{5V} + \frac{4\sqrt{\pi}}{5} \frac{\Lambda(T+1)^2}{V\sqrt{V}} \right\}. \tag{A.7}$$

For  $j \geq 1$  the summand is at most  $1/(5jV) + \sqrt{\pi}|t_j - 2\pi j|/(10j\sqrt{V})$ .

We now verify (5.11). Standardize by  $B$  and put  $d = T_X/B^2 \in [1/4, 2]$ . The Lyapunov ratio is  $d/B$ , and the support bounds satisfy the hypotheses of (A.3). Raw frequencies  $|u| \leq 1/128$  lie in that estimate's range; raw jitter width one corresponds to  $h = 1/d \leq 4$ . Its integral is at most  $300/B^2$ , and after (A.1) and multiplication by  $\sqrt{m}$  it costs at most  $400/\sqrt{m}$ . Equation (A.7) costs at most  $(1 + \log T)\{m^{-1/2} + \xi(T+1)^2\}$  in this normalization. Between raw frequencies  $1/128$  and

1/4, the product is bounded by  $e^{-B^2 u^2/3}$ . Outside the resonance intervals it is bounded by  $e^{-V/50}$ . The Gaussian comparison tail beginning at  $B/128$  is at most

$$(32854/B^2 + 1/192)e^{-B^2/32768}.$$

Their total contribution after smoothing and scaling is below  $60000(1 + \log T)\sqrt{m}e^{-m/262144}$ , using  $V \geq 3m/16$ . Finally the cutoff  $BT$  costs at most  $24\sqrt{m}/(BT) < 56/T$ ; the standardized signed comparison has Lipschitz constant below one. These are exactly the terms of (5.10).

## B Quantitative confinement of the original supports

This appendix proves Proposition 4.1. Its first two lemmas turn a small lattice-moment deficit and a small smoothing loss into explicit distances. The remaining argument applies them to the contact equations and records the constants needed for the theorem.

### B.1 The deficit in Esseen's moment inequality

We use a variance-weighted decomposition to quantify the deficit; compare He and Cheng [4, Lemma A.9] for stability of a single lattice law. Put  $c = c_E$ ,  $p = p_E$ , and  $\tau = \tau_E$  within this subsection. Let  $Y$  be centered, nondegenerate, and supported on a translate of  $h\mathbb{Z}$ . Its oriented deficit is

$$\delta_Y = \frac{c\mathbb{E}|Y|^3 - \mathbb{E}Y^3 - 3h\mathbb{E}Y^2}{h\mathbb{E}Y^2}.$$

Write  $P_-$  for the law of  $-Y$  on  $(0, \infty)$ ,  $P_+$  for the law of  $Y$  there, and  $m = \mathbb{E}Y_+ = \mathbb{E}Y_- > 0$ . The centered two-point law  $P_{u,v}$  at  $-u, v$  has masses  $v/(u+v), u/(u+v)$ . Direct integration gives

$$\text{Law}(Y) = \mathbb{P}(Y=0)\delta_0 + \int P_{u,v} \lambda(du, dv), \quad \lambda(du, dv) = \frac{u+v}{m} P_-(du) P_+(dv).$$

Its variance-weighted mixing measure  $\omega(du, dv) = uv\lambda(du, dv)/\mathbb{E}Y^2$  is a probability measure. Set  $n = (u+v)/h \in \{1, 2, \dots\}$  and  $a = u/(u+v)$ . The identity

$$c\{a^2 + (1-a)^2\} - (1-2a) = 3 + 2c(a-p)^2$$

gives the exact formula

$$\delta_Y = \int \{3(n-1) + 2cn(a-p)^2\} d\omega. \tag{B.1}$$

This proves the oriented inequality and, by reflection, Esseen's inequality (1.6), including its equality case.

If  $\delta_Y < 3$ , some component has  $n = 1$ . Its two points are the unique nearest lattice points straddling zero, say  $-ah, (1-a)h$ ; in particular this translate does not contain zero. Its variance weight is at least  $1 - \delta_Y/3$ . Every other component has variance at least  $h^2 a(1-a)$ , so this variance weight is at most its ordinary mixture mass. Thus, for a centered law  $Q$ ,

$$\begin{aligned} \text{Law}(Y) &= (1 - r_Y) \text{Law}(h(B_a - a)) + r_Y Q, \quad r_Y \leq \delta_Y/3, \\ |a - p| &\leq \sqrt{\frac{\delta_Y}{2c(1 - \delta_Y/3)}}, \quad h^2 a(1-a) \leq \mathbb{E}Y^2 \leq \frac{h^2 a(1-a)}{1 - \delta_Y/3}. \end{aligned} \tag{B.2}$$

If  $|Y| \leq M$ , coupling the mixture and then the two Bernoulli laws bounds their Wasserstein distance by

$$W_1(\text{Law}(Y), \text{Law}(h(B_p - p))) \leq \frac{2M}{3}\delta_Y + 2h\sqrt{\frac{\delta_Y}{2c(1 - \delta_Y/3)}}.$$

Here  $W_1$  is the infimum of  $\mathbb{E}|Y - Z|$  over couplings.

For a collection of centered  $h$ -lattice variables, use one common orientation and set

$$V = \sum_i \mathbb{E}Y_i^2, \quad B = \sum_i \mathbb{E}|Y_i|^3, \quad K = \sum_i \mathbb{E}Y_i^3, \quad \Delta_L = (cB - K - 3hV)/(hV).$$

Degenerate coordinates have zero variance weight. Applying the preceding bound on  $\delta_{Y_i} \leq 1$ , the crude bound  $M + h$  on its complement, and Cauchy–Schwarz gives

$$\frac{1}{V} \sum_i \mathbb{E}Y_i^2 W_1(\text{Law}(Y_i), \text{Law}(h(B_p - p))) \leq (5M/3 + h)\Delta_L + h\sqrt{3/c}\sqrt{\Delta_L}. \quad (\text{B.3})$$

There are also useful moment-ratio estimates. Put

$$a_D = \sqrt{(1 + D/3)D/(2c)}, \quad b_D = \tau D/3 + D/c + (2/c)a_D, \quad k_D = D/(3c) + 2a_D.$$

If  $\Delta_L \leq D$ , then

$$\left| \frac{B}{hV} - \tau \right| \leq b_D, \quad \left| \frac{K}{hV} - \frac{1}{c} \right| \leq k_D, \quad \frac{B}{hV} \geq \frac{1}{2}. \quad (\text{B.4})$$

Indeed (B.1) gives  $\mathbb{E}_\omega(n - 1) \leq D/3$ ,  $\mathbb{E}_\omega[n(a - p)^2] \leq D/(2c)$ , and  $|\mathbb{E}_\omega[n(a - p)]| \leq a_D$ . Expand  $n\{a^2 + (1 - a)^2\}$  and  $n(1 - 2a)$  about  $p$ . The last inequality follows directly from  $n \geq 1$ . All these formulas allow arbitrary variance weights, so they also hold when each sum is multiplied by a common positive factor.

## B.2 Rounding at an approximate period

For the selected array put  $Y_i = X_i/\ell$ . Then  $|Y_i| \leq 8$ ,  $\mathbb{E}Y_i^2 \leq 9$ , and

$$\ell^2 \sum_i \mathbb{E}Y_i^2 = \ell^2 \sum_i \mathbb{E}|Y_i|^3 = 1.$$

At a nonzero frequency  $u$ , take the phase of  $\mathbb{E}e^{iuY_i}$ , using any phase if the expectation vanishes. Round  $Y_i$  to the nearest point of the corresponding lattice of span  $h = 2\pi/|u|$  and then center the rounded variable, calling it  $Q_i$ . The inequality  $|v| \leq (\pi/2)|e^{iv} - 1|$  for  $|v| \leq \pi$  gives

$$e := \ell^2 \sum_i \mathbb{E}|Y_i - Q_i|^2 \leq h^2 g(u)/8, \quad |Q_i| \leq 8 + h. \quad (\text{B.5})$$

In detail the uncentered squared rounding error is at most  $(h^2/16)\mathbb{E}|e^{iuY_i} - e^{i\theta_i}|^2 = (h^2/8)(1 - |\mathbb{E}e^{iuY_i}|)$ , which is no larger than the expression using  $1 - |\mathbb{E}e^{iuY_i}|^2$ . Centering decreases that error. If  $M = 8 + h$ , Cauchy–Schwarz yields

$$\begin{aligned} \ell^2 \sum_i |\mathbb{E}Q_i^2 - \mathbb{E}Y_i^2| &\leq 2\sqrt{e} + e, \\ \ell^2 \sum_i |\mathbb{E}|Q_i|^3 - \mathbb{E}|Y_i|^3|, \quad \ell^2 \sum_i |\mathbb{E}Q_i^3 - \mathbb{E}Y_i^3| &\leq (3M/2)\sqrt{e}(2 + \sqrt{e}). \end{aligned} \quad (\text{B.6})$$

For the cubic bounds use  $||x|^3 - |y|^3|, |x^3 - y^3| \leq (3M/2)|x - y|(|x| + |y|)$ . These bounds depend on total variance, not on the number of coordinates.

### B.3 Comparing two small smoothing losses

**Lemma B.1.** *Let  $F$  be the CDF of a finite positive measure and let  $J = F * \text{Unif}[-h/2, h/2]$ ,  $h > 0$ . Suppose  $J(x)$  differs from an affine function of slope  $q > 0$  by at most  $\varepsilon$  on  $[-D_0 - 2h, 3h]$ . Put  $H(a) = J(a + h/2) - F(a)$ . If  $0 \leq d \leq D_0$  and  $H(0), H(-d) \leq s$ , then, for  $0 < r_0 \leq h/4$  and  $K_0 = \lceil D_0/h \rceil$ ,*

$$\text{dist}(d, h\mathbb{Z}) \leq 3r_0 + \frac{4s + (5 + 4K_0)\varepsilon + 4K_0h\varepsilon/r_0}{q}. \quad (\text{B.7})$$

*Proof.* For the measure  $\nu$  of  $F$ , write

$$w_a(t) = (1 - (t - a)/h)\mathbf{1}_{(a, a+h)}(t), \quad H(a) = \int w_a(t)\nu(dt) \geq 0.$$

For  $0 < d < h$ , define the nonnegative trapezoid

$$T_d(t) = \frac{1}{h} \int_{h/2-d}^{h/2-d/2} \mathbf{1}_{\{|x-t| < h/2\}} dx.$$

Its integral against  $\nu$  equals  $J(h/2 - d/2) - J(h/2 - d)$ . Splitting at  $-d, -d/2, 0, h - d, h - d/2, h$  gives the pointwise bound

$$\min\{1, 2(h - d)/d\} T_d(t) \leq w_0(t) + w_{-d}(t).$$

Integrating and applying the affine approximation to the two  $J$  values gives

$$q \min(d/2, h - d) \leq H(0) + H(-d) + 2\varepsilon. \quad (\text{B.8})$$

The pointwise bound also holds at the endpoints, with the open intervals in  $w_a$ , so atoms require no continuity assumption on  $F$ .

To transfer over several periods, let  $\bar{H}_{r_0}(a) = r_0^{-1} \int_a^{a+r_0} H(t)dt$ . Integrating the definition of the uniform convolution gives

$$\begin{aligned} \bar{H}_{r_0}(a + h) - \bar{H}_{r_0}(a) &= r_0^{-1} \int_a^{a+r_0} [J(t + 3h/2) - J(t + h/2)]dt \\ &\quad - (h/r_0)[J(a + r_0 + h/2) - J(a + h/2)]. \end{aligned}$$

The affine terms cancel, so its magnitude is at most  $2\varepsilon(1 + h/r_0)$ . Moreover  $H(a + t) \leq H(a) + (t/h)\nu((a + t, a + t + h))$  and the mass is at most  $2(qh + \varepsilon)$ , by the affine bound on  $J$ . Hence  $\bar{H}_{r_0}(a) \leq H(a) + qr_0 + \varepsilon r_0/h$ . Write  $d = kh + d_0$ ,  $0 \leq d_0 < h$ . Transfer from  $-d$  to  $-d_0$  using the preceding difference bound. If  $d_0$  is within  $2r_0$  of an endpoint, the conclusion is immediate. Otherwise choose a point in the averaging interval where  $H$  is at most its average and apply (B.8), using  $H(0) \leq s$ . The threshold displacement is at most  $r_0$ ; the resulting bound is (B.7). All arguments of  $J$  stay in the stated window.  $\square$

### B.4 The scale and the maximizing threshold

We prove Proposition 4.1. Retain the notation of Section 2 for the actual selected maximizer. For scaled coordinates write

$$Y_i = X_i/\ell, \quad v_i = \mathbb{E}Y_i^2, \quad M_i = \mathbb{E}[Y_i|Y_i|], \quad \theta = \ell^{-1} \sum_i \mathbb{E}X_i^3, \quad |\theta| \leq 1.$$

Let  $\rho, r, T, \varepsilon, h$  be as in Proposition 3.2. We give the error quantities explicitly so their later specialization requires no unspecified constants:

$$\begin{aligned}
u &= \rho + \ell/2, & D &= 16u + 256r, \\
H_h &= 72r + 5b_D, & Z^2 &= 10(u + 18r), \\
S_0 &= 2\rho + 18r + 460\ell^2, & A_0 &= 2\rho + 24\{4\ell^2 + (Z + 32\ell)^2/2 + \ell/5\}, \\
\gamma &= 300\sqrt{A_0} + 11S_0 + 131A_0, & W_0 &= 21D + 3\sqrt{D} + 42r, \\
E_P &= 14000(u + 18r + \eta) + 1100Z^2 + 4000\ell + 4000\ell^2 + 1100\eta, \\
E_{\text{end}} &= E_P + 6100\gamma + 700\gamma^2 + 500H_h.
\end{aligned} \tag{B.9}$$

Here  $a_D, b_D, k_D$  are those in (B.4). The proof below uses only

$$\begin{aligned}
r &\leq 10^{-6}, \quad D \leq 10^{-3}, \quad A_0 \leq 9/64, \quad \gamma \leq 1/100, \\
E_P &\leq 10^{-4}, \quad E_{\text{end}} \leq 1/2000, \quad W_0 \leq 1/4.
\end{aligned} \tag{B.10}$$

These will follow from (4.1) at the end.

The case  $h = 0$  would give  $R \leq \phi(0)/6 + \rho < C_E$  by (3.3), so  $h > 0$ . Initially  $h \leq 128\pi < 404$  and  $g(2\pi/h) \leq r^2/4$ . Equations (B.5)–(B.6) give  $\sqrt{e} \leq 101r$ ,  $|Q_i| \leq 412$ . If  $V_Q = \ell^2 \sum \mathbb{E}Q_i^2$  and  $B_Q = \ell^2 \sum \mathbb{E}|Q_i|^3$ , then  $V_Q \geq 1 - 202r$  and  $B_Q \leq 1 + 125000r$ . Since  $B_Q/(hV_Q) \geq 1/2$ , it follows that

$$h \leq 2(1 + 125000r)/(1 - 202r) < 5/2.$$

Apply the same rounding bounds with this improved span. Writing  $K_Q = \ell^2 \sum \mathbb{E}Q_i^3$ , we obtain

$$\sqrt{e} \leq r, \quad |V_Q - 1| \leq 3r, \quad |B_Q - 1| \leq 32r, \quad |K_Q - \theta| \leq 32r. \tag{B.11}$$

The lattice inequality in either orientation implies

$$3h + |\theta| \leq c + 256r. \tag{B.12}$$

The moment-error allowance is  $[32(c + 1) + 9h]r < 256r$ .

Unsmoothing at the actual threshold  $z$ , with shift  $h\ell/2$ , gives

$$R \leq \phi(z)\{h/2 + \theta(1 - z^2)/6\} + u. \tag{B.13}$$

The Gaussian increment has error at most  $\phi(0)h^2\ell^2/8$ , and the skew correction adds at most  $h\ell^2/18$ ; together they are below  $\ell^2/2$ . Since  $\phi(z)$  and  $|(1 - z^2)\phi(z)|$  are at most  $\phi(0)$ ,

$$3h + |\theta| > c - 16u, \quad R \leq C_E + u + 18r. \tag{B.14}$$

It follows that  $h > 3/2$ . Orient the rounded variables by sign  $\theta$ , arbitrarily if  $\theta = 0$ . Equations (B.11)–(B.14) bound their aggregate lattice deficit by  $D$ , its denominator  $hV_Q$  being greater than one. The moment-ratio estimates yield

$$|h - h_E| \leq H_h, \quad ||\theta| - 1/\sqrt{10}| \leq 34r + 3k_D + H_h/6. \tag{B.15}$$

For  $D \leq 10^{-3}$ ,  $a_D \leq \sqrt{D}/3$ ,  $b_D \leq D/2 + \sqrt{D}/9$ , and  $k_D \leq D/18 + 2\sqrt{D}/3$ . Thus  $H_h < 1/40$  and the second error is below  $1/10$ . Since  $31/16 < h_E < 41/21$ , this proves

$$19/10 < h < 2, \quad |\theta| > 1/5. \tag{B.16}$$

If  $\theta < 0$ , the envelope in (B.13) has its maximum at zero because  $h > |\theta|$ ; it would imply  $|\theta| < 128r + 8u = D/2$ , a contradiction. Hence  $\theta > 0$ . Finally  $C_E < (C_E + 18r)e^{-z^2/2} + u$ , so

$$|z| \leq Z, \quad 0 \leq K - C_E \leq u + 18r + \eta. \tag{B.17}$$

For this last bound use  $-\log(1 - x) \leq 2x$  for  $x \leq 1/2$ .

## B.5 Deletion with the same period and the support spacing

Deleting coordinate  $i$  subtracts at most  $\ell^2$  from  $g$ . Thus its high-frequency product bound is at most  $e^{1/2} < 2$  times the full-row bound. Keep the original  $h$  for every deletion. Write  $\sigma_i^2 = 1 - s_i \geq 3/4$ ,  $\kappa_- = \sum_{j \neq i} \mathbb{E} X_j^3$ , let  $J_i$  be the CDF of that deletion plus the same jitter, and put

$$G_i^*(x) = \Phi(x/\sigma_i) + \frac{\kappa_-}{6\sigma_i^3}(1 - (x/\sigma_i)^2)\phi(x/\sigma_i).$$

The logarithmic remainder in Appendix A remains  $(49/3)\ell^2 t^4$ . The low-frequency error is bounded by  $\ell^2 e^{-t^2/5} \{(49/3)t^4 + t^6/72\}$ , whose positive integral is below  $206\ell^2$ . The jitter-factor error is below  $\ell^2$ . For  $a = \delta/\ell$ , the Gaussian comparison tail is at most  $4/(3a^2) + \ell/(2a) < 5500\ell^2$ ; use  $\int_a^\infty t^2 e^{-3t^2/8} dt \leq 3/a$ . Also  $G_i^*$  is Lipschitz with constant below  $1/2$ . The weaker signed smoothing bound therefore gives

$$\|J_i - G_i^*\|_\infty \leq 6000\ell^2 + 24r\ell/\delta^2 + 2\log(T/\delta)e^{-\epsilon^2/(2\ell^2)} + 20\ell/T \leq 2\rho\ell. \quad (\text{B.18})$$

The last inequality uses the four allocations in (3.2).

At any original support point  $y = \ell t$ , the contact equality and (2.6) give

$$|G_i(z - y) - \Phi((z - y)/\sigma_i) - R\ell| \leq 450\ell^3. \quad (\text{B.19})$$

For the constant,  $|t| \leq 8$ ,  $v_i, |M_i| \leq 9$ ,  $\mathbb{E}|Y_i|^3 \leq 27$ , and  $K \leq 23/50$ . The Gaussian remainder contributes less than  $51\ell^3$ , the quadratic  $K$  term less than  $51\ell^3$ , and the remaining cubic, moment, and linear terms less than  $348\ell^3$ . Equations (B.12), (B.18), and (B.19) then imply

$$0 \leq [J_i(z - y + h\ell/2) - G_i(z - y)]/\ell \leq S_0. \quad (\text{B.20})$$

The scale corrections cost at most  $10\ell^2$  after division by  $\ell$ : use  $\sigma_i^{-1} - 1 \leq s_i$ ,  $\sigma_i^{-3} - 1 \leq 3s_i$ , and  $|\kappa_-/\ell| \leq \theta + 27\ell^2$ .

Fix a support point  $x$ . The function  $\ell^{-1}J_i(z - x + \ell a)$  differs from an affine function of slope  $\phi(0)$  by at most  $A_0$  on  $[-16 - 2h, 3h]$ . Indeed  $|a| \leq 24$ ,  $|z - x + \ell a| \leq Z + 32\ell$ , and

$$|(G_i^*)'(w) - \phi(0)| \leq 4\ell^2 + w^2/2 + \ell/5.$$

Integrate and use (B.18). Apply Lemma B.1 to the scaled positive measure, using (B.20),  $q = \phi(0)$ ,  $K_0 \leq 11$ , and  $r_0 = \sqrt{A_0} \leq h/4$ . Its bound is at most  $3\sqrt{A_0} + (8/3)(4S_0 + 49A_0 + 110\sqrt{A_0}) < \gamma$ . Thus for every pair of support points of each original coordinate,

$$\text{dist}((y - x)/\ell, h\mathbb{Z}) \leq \gamma. \quad (\text{B.21})$$

## B.6 The support polynomial and the two classes

Define

$$P_i(t) = t^3 - c|t|^3 + (3c/2)t^2 + 3(cM_i - v_i)t.$$

Every point of the full scaled support satisfies

$$P_i(t) \geq -E_P. \quad (\text{B.22})$$

To verify this with constants, divide (2.8) by  $\phi(0)\ell^3/6$  and use (2.7). Its Taylor remainder divided by  $\ell^3$  is below  $224\ell$ . Furthermore  $|\phi''(z) + \phi(0)| \leq (3/2)\phi(0)Z^2$ , since  $\|\phi''''\|_\infty \leq 3\phi(0)$  and  $\phi'''(0) = 0$ . The respective error bounds are

$$13184(K - C_E), \quad 1092Z^2, \quad 3584\ell, \quad 3500\ell^2, \quad 1008\eta.$$

For the first, use  $|-|t|^3 + (3/2)t^2 + 3M_i t| \leq 824$  and  $6/\phi(0) < 16$ . The last two come from  $6R\ell s_i^2 + \eta b_i + 4\eta \ell s_i$ . Equation (B.17) makes their sum at most  $E_P$ .

For a nondegenerate coordinate let  $-a, b$  be its support extremes. Centering gives  $v_i \leq ab$  and  $|M_i| \leq v_i$ . If  $a, b \geq 1/100$ , divide the endpoint inequalities by  $a, b$  and add. The linear terms cancel, giving

$$(c+1)a^2 + (c-1)b^2 \leq (3c/2)(a+b) + E_P(a+b)/(ab).$$

The left side is at least  $3(a+b)^2$ , because  $(c^2-1)/(2c) = 3$ . Therefore  $a+b \leq c/2 + E_P/(3ab) \leq 25/8 + 1/3 < 7/2$ . If  $a < 1/100$  and  $a+b \geq 7/2$ , then  $b > 3.49$  and

$$P_i(b) \leq b^2[-(c-1)b + 3c/2 + 3(c-1)a] < -1,$$

contradicting (B.22). If  $b < 1/100$ , use  $P_i(-a) \leq a^2[-(c+1)a + 3c/2 + 3(c+1)b] < -1$ . Every support diameter is consequently below  $7/2$ .

Partition coordinates according as their scaled support diameter is less than  $h/2$  or at least  $h/2$ . In the first class, (B.21) implies diameter at most  $\gamma$ , so centering gives  $|Y_i| \leq \gamma$ . In the second class the diameter differs from  $h$  by at most  $\gamma$ : zero and two multiples are excluded by (B.16) and the diameter bound. Every support point lies within  $2\gamma$  of an extreme. Split at the midpoint, use its upper-cluster indicator  $B_i$ , and take conditional means. This gives exactly

$$Y_i = h_i(B_i - p_i) + U_i, \quad \mathbb{E}(U_i | B_i) = 0, \quad |h_i - h| \leq 5\gamma, \quad |U_i| \leq 2\gamma. \quad (\text{B.23})$$

Both cluster probabilities are positive, because both extremes are support points and the clusters are separated.

We next bound those probabilities away from zero and one. On  $[-8, 8]$ ,  $|P'_i| < 1800$ . Moving an extreme to its cluster mean costs at most  $3600\gamma$ . Conditional centering gives  $|M_i - M_i^0| \leq \mathbb{E}U_i^2 \leq 4\gamma^2$  and  $v_i - v_i^0 = \mathbb{E}U_i^2 \leq 4\gamma^2$ , where the superscript denotes the pure two-point law. The first bound uses the Lipschitz derivative of  $x|x|$  and also holds if a cluster crosses zero. Their polynomial contribution is below  $700\gamma^2$ . At a pure-law endpoint the polynomial is  $h_i^3 C_3(p_i) + h_i^2 C_2(p_i)$ , with  $|C_3| < 29$ ,  $|C_2| < 75/8$ . Its gap derivative for gaps at most  $9/4$  is below  $500$ . Thus moving to the ideal gap  $h_E$  leaves both endpoint values at least  $-E_{\text{end}}$ .

Put  $t_0 = 3\sqrt{10}/2$ . At its upper and lower endpoints those ideal-gap values are

$$h_E^3(1-p)^2 A(p), \quad h_E^3 p^2 B(p),$$

where

$$\begin{aligned} A(p) &= t_0 - (c-1) + 4(c-1)p - 6cp^2, \\ B(p) &= t_0 - 3(c-1) + (8c-4)p - 6cp^2. \end{aligned}$$

For  $p \leq 1/4$ ,  $B$  is increasing and  $B(p) \leq (\sqrt{10}-17)/8 < -1$ . If  $p \in [1/100, 1/4]$ , the lower endpoint is below  $-6/10000 < -1/2000$ , since  $h_E^3 > 6$ . For  $p \leq 1/100$ , use  $A(0) = -p_E < -2/5$ ,  $A' \leq 21$ , so  $A(p) < -19/100$ . For  $p \geq 3/4$ ,  $A$  is decreasing and  $A(p) \leq (\sqrt{10}-49)/8 < -5$ , excluding  $[3/4, 99/100]$ . For  $p \geq 99/100$ ,  $B(1) < -2$  and  $|B'| \leq 46$  give  $B(p) < -1$ . In view of  $E_{\text{end}} \leq 1/2000$ , all these cases are impossible. Hence every two-cluster probability lies in  $[1/4, 3/4]$ .

## B.7 The extra coordinates and the parameter check

Lattice stability and the rounding coupling give

$$\sum_i \ell^2 v_i W_1(\text{Law}(Y_i), \text{Law}(h(B_{p_E} - p_E))) \leq W_0. \quad (\text{B.24})$$

For the rounded variables (B.3) is at most  $(1 + 3r)(20D + 2\sqrt{D}) \leq 21D + 3\sqrt{D}$ . The change of variance weights has total absolute mass at most  $3r$ , and the distances are at most 13, costing  $39r$ . The original variance-weighted rounding distance is at most  $3r$ , by  $v_i \leq 9$  and the squared error bound  $e \leq r^2$ . This proves (B.24) with the  $42r$  allowance in  $W_0$ .

The target law has absolute first moment  $2hp_{\text{EQE}} > 9/10$ . On the small-diameter class  $|Y_i| \leq \gamma$ . Its total original variance  $V_s$  is therefore at most  $W_0/(9/10 - \gamma) \leq 2W_0$ . Each remaining coordinate has scaled variance between  $1/2$  and  $2$ , by (B.16), (B.23), and  $p_i \in [1/4, 3/4]$ . Their number  $m$  satisfies

$$\frac{1 - 2W_0}{2\ell^2} \leq m \leq \frac{2}{\ell^2}, \quad m \geq 1/(4\ell^2). \quad (\text{B.25})$$

Calling the small scaled coordinates  $Z_j$ ,

$$\frac{\sum_j \text{Var } Z_j}{mh^2} \leq \frac{4W_0}{h^2(1 - 2W_0)} < 3W_0.$$

The relative gap errors in (B.23) are below  $3\gamma$ , the residuals satisfy  $|U_i|/h < 2\gamma$ , and  $|Z_j|/h < \gamma$ . Thus the conclusion follows once  $3\gamma, 3W_0 \leq \lambda$  and  $\ell \leq 1/(2\sqrt{N})$ .

We finish by checking these conditions and (B.10). From (3.2),  $\ell \leq \rho/48000$  and  $r = \rho/196608$ . For  $\rho \leq 10^{-6}$ ,

$$\begin{aligned} D &\leq 17\rho, \quad Z^2 \leq 20\rho, \quad H_h \leq 43\rho + 3\sqrt{\rho}, \quad S_0 \leq 3\rho, \quad A_0 \leq 500\rho, \\ \gamma &\leq 7000\sqrt{\rho}, \quad W_0 \leq 14\sqrt{\rho}, \quad E_P \leq 51000\rho + 15100\eta. \end{aligned}$$

For  $\rho \leq 10^{-12}$  these imply

$$E_{\text{end}} \leq 5 \cdot 10^7 \sqrt{\rho} + 2 \cdot 10^4 \eta.$$

Before enlargement its  $\rho$  and  $\sqrt{\rho}$  coefficients are at most 34300072500 and 42701500, which proves this last bound. Substitute  $\rho = 10^{-26}\lambda^2$ ,  $\eta = 10^{-10}\lambda$ :

$$E_{\text{end}} \leq 7 \cdot 10^{-6}\lambda < 1/2000, \quad \gamma \leq 7 \cdot 10^{-10}\lambda, \quad W_0 \leq 1.4 \cdot 10^{-12}\lambda.$$

All the stated conditions follow, including  $E_P \leq 10^{-4}$ . The representations and the variance estimates concern the original coordinates throughout, proving Proposition 4.1.

## C Conditional moments and the one-sided comparison

This appendix proves the estimates used in Section 5. All conditioning is on the actual Bernoulli count. Dependence between a label and its residual is preserved.

### C.1 Exponential tilting and projection

Retain  $p_i \in [1/4, 3/4]$ ,  $v_i = p_i(1 - p_i)$ ,  $\sum_i v_i a_i = 0$ , and  $\Lambda_A = \sum_i v_i a_i^2$ . Suppose  $m \geq 10^{12}$  and  $|k - \mu| \leq 6\sqrt{m}$ . Tilt by  $e^{\theta K_0}$  so that the count mean is  $k$ , putting

$$r_i = \frac{p_i e^\theta}{1 - p_i + p_i e^\theta}, \quad w_i = r_i(1 - r_i), \quad V_\theta = \sum_i w_i.$$

For  $|\theta| \leq \log 2$ ,  $r_i \in [1/8, 7/8]$  and  $w_i \geq 7/64 > 1/10$ . The count mean changes by more than  $6\sqrt{m}$  in both directions before that boundary, so the tilt exists and

$$|\theta| \leq 60/\sqrt{m}, \quad |w_i/v_i - 1| \leq 2|\theta|, \quad |r_i - p_i - \theta v_i| \leq \theta^2 v_i. \quad (\text{C.1})$$

The conditional law given  $K_0 = k$  is unchanged. Define

$$c_\theta = \frac{\sum_i w_i a_i}{V_\theta}, \quad b_i = a_i - c_\theta, \quad L = \sum_i b_i (B_i - r_i), \quad Q = \sum_i b_i^2, \quad Q_\theta = \sum_i w_i b_i^2.$$

On the count event,  $A = m_k^0 + L$ , where

$$|m_k^0| = \left| \sum_i a_i (r_i - p_i) \right| \leq 1800\sqrt{\Lambda_A}/\sqrt{m}, \quad |Q_\theta - \Lambda_A| \leq 180\Lambda_A/\sqrt{m}, \quad Q \leq 16\Lambda_A. \quad (\text{C.2})$$

For the second bound expand  $Q_\theta = \sum_i w_i a_i^2 - (\sum_i w_i a_i)^2/V_\theta$ . The errors are at most  $2|\theta|\Lambda_A$  and  $8\theta^2\Lambda_A$ , together at most  $3|\theta|\Lambda_A$ . Also  $\Lambda_A/2 \leq Q_\theta \leq 3\Lambda_A/2$ ; using  $w_i \geq 7/64$  proves the last bound. If  $\Lambda_A = 0$ , all the gap estimates are immediate, so suppose it is positive.

## C.2 Fourier moments with their residual factors

For  $|t| \leq 1/4$ , put  $z_i(t) = r_i e^{it}/(1 - r_i + r_i e^{it})$ . The first centered complex cumulant is  $c_{i,1} = z_i - r_i$ ; the next three are

$$c_{i,2} = z_i(1 - z_i), \quad c_{i,3} = z_i(1 - z_i)(1 - 2z_i), \quad c_{i,4} = z_i(1 - z_i)(1 - 6z_i(1 - z_i)).$$

Both  $z_i$  and  $1 - z_i$  have modulus at most two. Differentiation gives

$$|c_{i,1} - i w_i t| \leq 10t^2, \quad |c_{i,2} - w_i| \leq 20|t|, \quad |c_{i,3}| \leq 20, \quad |c_{i,4}| \leq 100.$$

For  $M_j = \sum_i b_i^j c_{i,j}$ , the exact cancellation  $\sum_i w_i b_i = 0$  yields

$$\begin{aligned} |M_1| &\leq 10\sqrt{mQ}t^2, \quad |M_2 - Q_\theta| \leq 20Q|t|, \quad |M_2| \leq 6Q, \\ |M_3| &\leq 20Q^{3/2}, \quad |M_4| \leq 100Q^2. \end{aligned} \quad (\text{C.3})$$

The tilted count transform  $F_\theta$  satisfies  $|F_\theta(t)| \leq e^{-mt^2/50}$  on  $[-\pi, \pi]$ . Lemma A.1 gives  $\mathbb{P}_\theta(K_0 = k) \geq 1/(3\sqrt{m})$ . After division by this probability, upper bounds for the Gaussian integrals of powers 0, 1, 2, 4, 8 are respectively

$$7, \quad 25m^{-1/2}, \quad 175m^{-1}, \quad 13125m^{-2}, \quad 3 \cdot 10^8 m^{-4}. \quad (\text{C.4})$$

These follow by scaling  $t\sqrt{m}$  in  $(2\pi)^{-1} \int e^{-mt^2/50} |t|^j dt$ . On  $1/4 \leq |t| \leq \pi$ , expand the moment before factoring the product. A term of degree  $j \leq 4$  removes at most four Bernoulli factors; the remaining product is at most  $e^{-m/810}$ . The conditional tail contribution is therefore at most  $3m^{(j+1)/2} Q^{j/2} e^{-m/810}$ . For  $j = 1, 2, 4$  and  $m \geq 10^8$ , these are at most  $\sqrt{Q}/\sqrt{m}$ ,  $Q/\sqrt{m}$ , and  $Q^2$ . For instance  $e^{-m/810} \leq 10^{18}/m^5$  reduces the verification to decreasing powers of  $m$ .

The first, second, and fourth moment numerators are exactly

$$F_\theta M_1, \quad F_\theta(M_1^2 + M_2), \quad F_\theta(M_1^4 + 6M_1^2 M_2 + 3M_2^2 + 4M_1 M_3 + M_4).$$

Equations (C.3)–(C.4), with the tails just bounded, imply

$$\begin{aligned} |\mathbb{E}(L \mid K_0 = k)| &\leq 2000\sqrt{Q}/\sqrt{m}, \\ |\text{Var}(L \mid K_0 = k) - Q_\theta| &\leq 2000Q/\sqrt{m}, \quad \mathbb{E}(L^4 \mid K_0 = k) \leq 2000Q^2. \end{aligned} \quad (\text{C.5})$$

Here is numerical detail for these enlargements. The mean's central coefficient is 1750. The second moment errors are at most  $500Q/\sqrt{m} + 1312500Q/m + 2Q/\sqrt{m}$ , and the squared mean costs at most  $4 \cdot 10^6 Q/m$ . The five fourth-moment central coefficients, in the displayed order, are at most

$$3 \cdot 10^{12}/m^2, \quad 47250000/m, \quad 756, \quad 140000/\sqrt{m}, \quad 700,$$

all multiplying  $Q^2$ . Their sum and the tail are below  $2000Q^2$ . Centering raises the fourth-moment bound to  $20000Q^2$ . Combining (C.2) and (C.5) proves (5.5)–(5.6).

For completeness, the relative label estimate (5.7) also follows numerically. Under the tilt, deleting a Bernoulli changes count variance by at most  $1/4$  and displaces the new threshold from its mean by at most one. The Gaussian mass ratio has logarithm bounded by  $1/V_\theta$ . The  $6/V$  errors and a lower bound by half the centered Gaussian mass then give a deleted-count ratio error at most

$$4/V_\theta + 28/(\phi(0)\sqrt{V_\theta}) \leq 300/\sqrt{m}.$$

Each tilted label probability differs relatively from its original probability by at most  $2|\theta| \leq 120/\sqrt{m}$ . Multiplication gives  $500/\sqrt{m}$  for  $m \geq 10^{12}$ . This is a relative bound, so arbitrarily small within-cluster noise variances remain controlled after conditioning.

### C.3 Proof of the one-sided loss

We prove (5.14), following the clipping method of He and Cheng [4, Lemma 3.2 and equation (83)]. First suppose  $V_0$  is centered and  $|V_0| \leq a = 1/10$ . For  $u \geq 0$ ,  $\mathbb{E}(V_0)_+ \leq u + a\mathbb{P}(V_0 > u)$  and  $f(u) \geq 3u/4$  give  $\mathbb{P}(V_0 > u) + f(u) \geq 3\mathbb{E}|V_0|/8$ . If  $u = -r$ ,  $0 < r \leq a$ , put  $t = \mathbb{P}(V_0 > -r)$ . Centering gives  $0 \leq at - r(1 - t)$ , hence  $r \leq 2at$ . Also  $\mathbb{E}|V_0| = 2\mathbb{E}(V_0)_+ \leq 2at$ , so  $t + f(-r) \geq t - 5r/4 \geq 3t/4 \geq 3\mathbb{E}|V_0|/8$ . For  $a < r \leq 13/20$ , the probability is one and  $1 + f(-r) \geq 3/16 \geq 3\mathbb{E}|V_0|/8$ .

For general centered  $Y$ , clip at  $1/20$ , obtaining  $T$ , and put  $b = \mathbb{E}T$ ,  $V_0 = T - b$ ,  $M_4 = \mathbb{E}Y^4$ . Then

$$|b| \leq 8000M_4, \quad \mathbb{P}(T \neq Y) \leq 160000M_4, \quad \mathbb{E}|V_0| \geq \mathbb{E}|Y| - 16000M_4.$$

If  $M_4 \leq 1/160000$ , then  $|b| \leq 1/20$ ,  $|V_0| \leq 1/10$ , and  $|u - b| \leq 13/20$ . Apply the bounded case at  $u - b$ , and compare  $f(u - b)$  with  $f(u)$ . The total error coefficient is  $(3/8)16000 + 160000 + (5/4)8000 = 176000$ . This proves the first inequality in (5.14). Apply it to  $-Y$  and  $-f(-\cdot)$  for the second. All events in this proof are strict; atoms at the threshold are included with the conventions needed in (5.16).

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